

DPP-10

- Q.1 If $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar & $\vec{p}, \vec{q}, \vec{r}$ are reciprocal vectors to $\vec{a}, \vec{b}$ & $\vec{c}$ respectively, then $(\ell\vec{a} + m\vec{b} + n\vec{c}) \cdot (\ell\vec{p} + m\vec{q} + n\vec{r})$ is equal to : (where ℓ, m, n are scalars)
(A) $\ell^2 + m^2 + n^2$ (B) $\ell m + m n + n \ell$ (C) 0 (D) none of these
- Q.2 If $\vec{A}, \vec{B}$ & $\vec{C}$ are three non-coplanar vectors, then $(\vec{A} + \vec{B} + \vec{C}) \cdot [(\vec{A} + \vec{B}) \times (\vec{A} + \vec{C})]$ equals
(A) 0 (B) $[\vec{A} \vec{B} \vec{C}]$ (C) $2 [\vec{A} \vec{B} \vec{C}]$ (D) $- [\vec{A} \vec{B} \vec{C}]$
- Q.3 A plane P_1 has the equation $2x - y + z = 4$ and the plane P_2 has the equation $x + ny + 2z = 11$. If the angle between P_1 and P_2 is $\frac{\pi}{3}$ then the value(s) of 'n' is (are)
(A) $7/2$ (B) $17, -1$ (C) $-17, 1$ (D) $-7/2$
- Q.4 The three vectors $\hat{i} + \hat{j}, \hat{j} + \hat{k}, \hat{k} + \hat{i}$ taken two at a time form three planes. The three unit vectors drawn perpendicular to these three planes form a parallelopiped of volume :
(A) $1/3$ (B) 4 (C) $3\sqrt{3}/4$ (D) $4/3\sqrt{3}$
- Q.5 If $\vec{x}$ & $\vec{y}$ are two non collinear vectors and a, b, c represent the sides of a ΔABC satisfying $(a - b)\vec{x} + (b - c)\vec{y} + (c - a)(\vec{x} \times \vec{y}) = 0$ then ΔABC is
(A) an acute angle triangle (B) an obtuse angle triangle
(C) a right angle triangle (D) a scalene triangle
- Q.6 Given three non-zero, non-coplanar vectors $\vec{a}, \vec{b}, \vec{c}$ and $\vec{r}_1 = p\vec{a} + q\vec{b} + \vec{c}$ and $\vec{r}_2 = \vec{a} + p\vec{b} + q\vec{c}$ if the vectors $\vec{r}_1 + 2\vec{r}_2$ and $2\vec{r}_1 + \vec{r}_2$ are collinear then (p, q) is
(A) $(0, 0)$ (B) $(1, -1)$ (C) $(-1, 1)$ (D) $(1, 1)$
- Q.7 If the vectors $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar and ℓ, m, n are distinct scalars, then $\left[(\ell\vec{a} + m\vec{b} + n\vec{c}) (\ell\vec{b} + m\vec{c} + n\vec{a}) (\ell\vec{c} + m\vec{a} + n\vec{b}) \right] = 0$ implies :
(A) $\ell m + m n + n \ell = 0$ (B) $\ell + m + n = 0$
(C) $\ell^2 + m^2 + n^2 = 0$ (D) $\ell^3 + m^3 + n^3 = 0$
- Q.8 Let $\vec{r}_1, \vec{r}_2, \vec{r}_3, \dots, \vec{r}_n$ be the position vectors of points $P_1, P_2, P_3, \dots, P_n$ relative to the origin O. If the vector equation $a_1\vec{r}_1 + a_2\vec{r}_2 + \dots + a_n\vec{r}_n = 0$ holds, then a similar equation will also hold w.r.t. to any other origin provided
(A) $a_1 + a_2 + \dots + a_n = n$ (B) $a_1 + a_2 + \dots + a_n = 1$
(C) $a_1 + a_2 + \dots + a_n = 0$ (D) none
- Q.9 The orthogonal projection A' of the point A with position vector $(1, 2, 3)$ on the plane $3x - y + 4z = 0$ is
(A) $(-1, 3, -1)$ (B) $\left(-\frac{1}{2}, \frac{5}{2}, 1\right)$ (C) $\left(\frac{1}{2}, -\frac{5}{2}, -1\right)$ (D) $(6, -7, -5)$

Paragraph for Question Nos. 10 to 11

Consider a plane

$$x + y - z = 1 \text{ and the point } A(1, 2, -3)$$

A line L has the equation

$$x = 1 + 3r$$

$$y = 2 - r$$

$$z = 3 + 4r$$

- Q.10 The co-ordinate of a point B of line L, such that AB is parallel to the plane, is
(A) 10, -1, 15 (B) -5, 4, -5 (C) 4, 1, 7 (D) -8, 5, -9
- Q.11 Equation of the plane containing the line L and the point A has the equation
(A) $x - 3y + 5 = 0$ (B) $x + 3y - 7 = 0$ (C) $3x - y - 1 = 0$ (D) $3x + y - 5 = 0$

Paragraph for Question Nos. 12 to 15

Consider a triangular pyramid ABCD the position vectors of whose angular points are $A(3, 0, 1)$; $B(-1, 4, 1)$; $C(5, 2, 3)$ and $D(0, -5, 4)$. Let G be the point of intersection of the medians of the triangle BCD.

- Q.12 The length of the vector $\overrightarrow{AG}$ is
(A) $\sqrt{17}$ (B) $\sqrt{51}/3$ (C) $\sqrt{51}/9$ (D) $\sqrt{59}/4$
- Q.13 Area of the triangle ABC in sq. units is
(A) 24 (B) $8\sqrt{6}$ (C) $4\sqrt{6}$ (D) none
- Q.14 The length of the perpendicular from the vertex D on the opposite face is
(A) $14/\sqrt{6}$ (B) $2/\sqrt{6}$ (C) $3/\sqrt{6}$ (D) none
- Q.15 Equation of the plane ABC is
(A) $x + y + 2z = 5$ (B) $x - y - 2z = 1$ (C) $2x + y - 2z = 4$ (D) $x + y - 2z = 1$

Paragraph for Question Nos. 16 to 18

The equation of line: $\frac{x - x'}{a'} = \frac{y - y'}{b'} = \frac{z - z'}{c'}$

The equation of plane : $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$

Equation of plane through the intersection of the two planes

$$a_1x + b_1y + c_1z + d_1 = 0 \quad \text{and} \quad a_2x + b_2y + c_2z + d_2 = 0 : \\ (a_1x + b_1y + c_1z + d_1) + k(a_2x + b_2y + c_2z + d_2) = 0$$

- Q.16 The distance of the point $(1, -2, 3)$ from the plane $x - y + z = 5$ measured parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z-3}{-4}$ is
(A) $\sqrt{21}/5$ (B) $\sqrt{29}/5$ (C) $\sqrt{13}/5$ (D) $2/\sqrt{5}$
- Q.17 The equation of the plane through $(0, 2, 4)$ and containing the line $\frac{x+3}{3} = \frac{y-1}{4} = \frac{z-2}{-2}$ is
(A) $x - 2y + 4z - 12 = 0$ (B) $5x + y + 9z - 38 = 0$
(C) $10x - 12y - 9z + 60 = 0$ (D) $7x + 5y - 3z + 2 = 0$
- Q.18 The plane $x - y - z = 2$ is rotated through 90° about its line of intersection with the plane $x + 2y + z = 2$. Then equation of this plane in new position is
(A) $5x + 4y + z - 10 = 0$ (B) $4x + 5y - 3z = 0$
(C) $2x + y + 2z = 9$ (D) $3x + 4y - 5z = 9$

Paragraph for Question Nos. 19 to 21

Consider the three vectors $\vec{p}, \vec{q}$ and $\vec{r}$ such that

$$\vec{p} = \hat{i} + \hat{j} + \hat{k}; \quad \vec{q} = \hat{i} - \hat{j} + \hat{k}$$

$$\vec{p} \times \vec{r} = \vec{q} + c\vec{p} \text{ and } \vec{p} \cdot \vec{r} = 2$$

Q.19 The value of $[\vec{p} \ \vec{q} \ \vec{r}]$ is

- (A) $-\frac{5\sqrt{2}c}{|\vec{r}|}$ (B) $-\frac{8}{3}$ (C) 0 (D) greater than zero

Q.20 If $\vec{x}$ is a vector such that $[\vec{p} \ \vec{q} \ \vec{r}] \vec{x} = (\vec{p} \times \vec{q}) \times \vec{r}$, then $\vec{x}$ is

- (A) $c(\hat{i} - 2\hat{j} + \hat{k})$ (B) a unit vector
- (C) indeterminate, as $[\vec{p} \ \vec{q} \ \vec{r}]$ (D) $-\frac{1}{2}(\hat{i} - 2\hat{j} + \hat{k})$

Q.21 If $\vec{y}$ is a vector satisfying $(1 + c)\vec{y} = \vec{p} \times (\vec{q} \times \vec{r})$ then the vectors $\vec{x}, \vec{y}, \vec{r}$

- (A) are collinear
- (B) are coplanar
- (C) represent the coterminus edges of a tetrahedron whose volume is c cubic units.
- (D) represent the coterminus edges of a parallelepiped whose volume is c cubic units

[REASONING TYPE]

Q.22 Given lines $\frac{x-4}{2} = \frac{y+5}{4} = \frac{z-1}{-3}$ and $\frac{x-2}{1} = \frac{y+1}{3} = \frac{z}{2}$

Statement-1: The lines intersect.

because

Statement-2: They are not parallel.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
- (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
- (C) Statement-1 is true, statement-2 is false.
- (D) Statement-1 is false, statement-2 is true.

Q.23 Consider three vectors $\vec{a}, \vec{b}$ and $\vec{c}$

Statement-1: $\vec{a} \times \vec{b} = ((\hat{i} \times \vec{a}) \cdot \vec{b})\hat{i} + ((\hat{j} \times \vec{a}) \cdot \vec{b})\hat{j} + ((\hat{k} \times \vec{a}) \cdot \vec{b})\hat{k}$

because

Statement-2: $\vec{c} = (\hat{i} \cdot \vec{c})\hat{i} + (\hat{j} \cdot \vec{c})\hat{j} + (\hat{k} \cdot \vec{c})\hat{k}$

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
- (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
- (C) Statement-1 is true, statement-2 is false.
- (D) Statement-1 is false, statement-2 is true.

[MULTIPLE OBJECTIVE TYPE]

Select the correct alternative(s): (More than one are correct)

- Q.24 If $A(\bar{a})$; $B(\bar{b})$; $C(\bar{c})$ and $D(\bar{d})$ are four points such that
 $\bar{a} = -2\hat{i} + 4\hat{j} + 3\hat{k}$; $\bar{b} = 2\hat{i} - 8\hat{j}$; $\bar{c} = \hat{i} - 3\hat{j} + 5\hat{k}$; $\bar{d} = 4\hat{i} + \hat{j} - 7\hat{k}$
 d is the shortest distance between the lines AB and CD , then which of the following is True?
- (A) $d = 0$, hence AB and CD intersect (B) $d = \frac{[\vec{AB} \vec{CD} \vec{BD}]}{|\vec{AB} \times \vec{CD}|}$
- (C) AB and CD are skew lines and $d = \frac{23}{13}$ (D) $d = \frac{[\vec{AB} \vec{CD} \vec{AC}]}{|\vec{AB} \times \vec{CD}|}$
- Q.25 Consider four points $A(\bar{a})$; $B(\bar{b})$; $C(\bar{c})$ and $D(\bar{d})$, such that
 $\vec{GA} + \vec{GB} + \vec{GC} + \vec{GD} = \vec{0}$ for a point $G(\bar{g})$, if
- (A) G is the centroid of the tetrahedron $ABCD$
 (B) G lies on the line joining each of A, B, C, D to the centroid of the triangle formed by the other three
 (C) p.v. of G is collinear with the p.v. of the centroids of the triangle formed by any three of the four given points.
 (D) $\square ABCD$ is parallelogram with G as the point of intersection of the diagonals AC and BD .
- Q.26 Given the equations of the line $3x - y + z + 1 = 0$, $5x + y + 3z = 0$.
 Then which of the following is correct ?
- (A) symmetrical form of the equations of line is $\frac{x}{2} = \frac{y - \frac{1}{8}}{-1} = \frac{z + \frac{5}{8}}{1}$
- (B) symmetrical form of the equations of line is $\frac{x + \frac{1}{8}}{1} = \frac{y - \frac{5}{8}}{1} = \frac{z}{-2}$
- (C) equation of the plane through $(2, 1, 4)$ and perpendicular to the given lines is $2x - y + z - 7 = 0$
- (D) equation of the plane through $(2, 1, 4)$ and perpendicular to the given lines is $x + y - 2z + 5 = 0$
- Q.27 Given three vectors
 $\vec{U} = 2\hat{i} + 3\hat{j} - 6\hat{k}$; $\vec{V} = 6\hat{i} + 2\hat{j} + 3\hat{k}$; $\vec{W} = 3\hat{i} - 6\hat{j} - 2\hat{k}$
 Which of the following hold good for the vectors $\vec{U}$, $\vec{V}$ and $\vec{W}$?
- (A) $\vec{U}$, $\vec{V}$ and $\vec{W}$ are linearly dependent
 (B) $(\vec{U} \times \vec{V}) \times \vec{W} = \vec{0}$
 (C) $\vec{U}$, $\vec{V}$ and $\vec{W}$ form a triplet of mutually perpendicular vectors
 (D) $\vec{U} \times (\vec{V} \times \vec{W}) = \vec{0}$
- Q.28 Consider the family of planes $x + y + z = c$ where c is a parameter intersecting the coordinate axes at P, Q, R and α, β, γ are the angles made by each member of this family with positive x, y and z axis. Which of the following interpretations hold good for this family.
- (A) each member of this family is equally inclined with the coordinate axes.
 (B) $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 1$
 (C) $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 2$
 (D) for $c = 3$ area of the triangle PQR is $3\sqrt{3}$ sq. units.

[MATCH THE COLUMN]

Q.29

Column-I

Column-II

- | | |
|--|---|
| <p>(A) <i>Centre of the parallelopiped whose 3 coterminous edges $\vec{OA}$, $\vec{OB}$ and $\vec{OC}$ have position vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ respectively where O is the origin, is</i></p> | <p>(P) $\vec{a} + \vec{b} + \vec{c}$</p> |
| <p>(B) <i>$OABC$ is a tetrahedron where O is the origin. Positions vectors of its angular points A, B and C are $\vec{a}, \vec{b}$ and $\vec{c}$ respectively. Segments joining each vertex with the centroid of the opposite face are concurrent at a point P whose p.v.'s are</i></p> | <p>(Q) $\frac{\vec{a} + \vec{b} + \vec{c}}{3}$</p> |
| <p>(C) <i>Let ABC be a triangle the position vectors of its angular points are $\vec{a}, \vec{b}$ and $\vec{c}$ respectively. If $\vec{a} - \vec{b} = \vec{b} - \vec{c} = \vec{c} - \vec{a}$ then the p.v. of the orthocentre of the triangle is</i></p> | <p>(R) $\frac{\vec{a} + \vec{b} + \vec{c}}{4}$</p> |
| <p>(D) <i>Let $\vec{a}, \vec{b}, \vec{c}$ be 3 mutually perpendicular vectors of the same magnitude. If an unknown vector $\vec{x}$ satisfies the equation $\vec{a} \times (\vec{x} - \vec{b}) \times \vec{a} + \vec{b} \times (\vec{x} - \vec{c}) \times \vec{b} + \vec{c} \times (\vec{x} - \vec{a}) \times \vec{c} = 0$. Then $\vec{x}$ is given by</i></p> | <p>(S) $\frac{\vec{a} + \vec{b} + \vec{c}}{2}$</p> |

Q.30

Column-I

Column-II

- | | |
|---|--|
| <p>(A) Let O be an interior point of ΔABC such that $\vec{OA} + 2\vec{OB} + 3\vec{OC} = \vec{0}$, then the ratio of the area of ΔABC to the area of ΔAOC, is with O is the origin</p> | <p>(P) 0</p> |
| <p>(B) Let ABC be a triangle whose centroid is G, orthocentre is H and circumcentre is the origin 'O'. If D is any point in the plane of the triangle such that no three of O, A, B, C and D are collinear satisfying the relation $\vec{AD} + \vec{BD} + \vec{CH} + 3\vec{HG} = \lambda \vec{HD}$ then the value of the scalar 'λ' is</p> | <p>(Q) 1</p> <p>(R) 2</p> <p>(S) 3</p> |
| <p>(C) If $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are non zero vectors such that no three of them are in the same plane and no two are orthogonal then the value of the scalar $\frac{(\vec{b} \times \vec{c}) \cdot (\vec{a} \times \vec{d}) + (\vec{c} \times \vec{a}) \cdot (\vec{b} \times \vec{d})}{(\vec{a} \times \vec{b}) \cdot (\vec{d} \times \vec{c})}$ is</p> | |

[SUBJECTIVE TYPE]

- Q.31 If the lattice point $P(x, y, z)$, $x, y, z \in I$ with the largest value of z such that the P lies on the planes $7x + 6y + 2z = 272$ and $x - y + z = 16$ (given $x, y, z > 0$), find the value of $(x + y + z)$.
- Q.32 Given $\vec{A} = 2\hat{i} + 3\hat{j} + 6\hat{k}$, $\vec{B} = \hat{i} + \hat{j} - 2\hat{k}$ and $\vec{C} = \hat{i} + 2\hat{j} + \hat{k}$.
Compute the value of $|\vec{A} \times (\vec{A} \times (\vec{A} \times \vec{B})) \cdot \vec{C}|$.